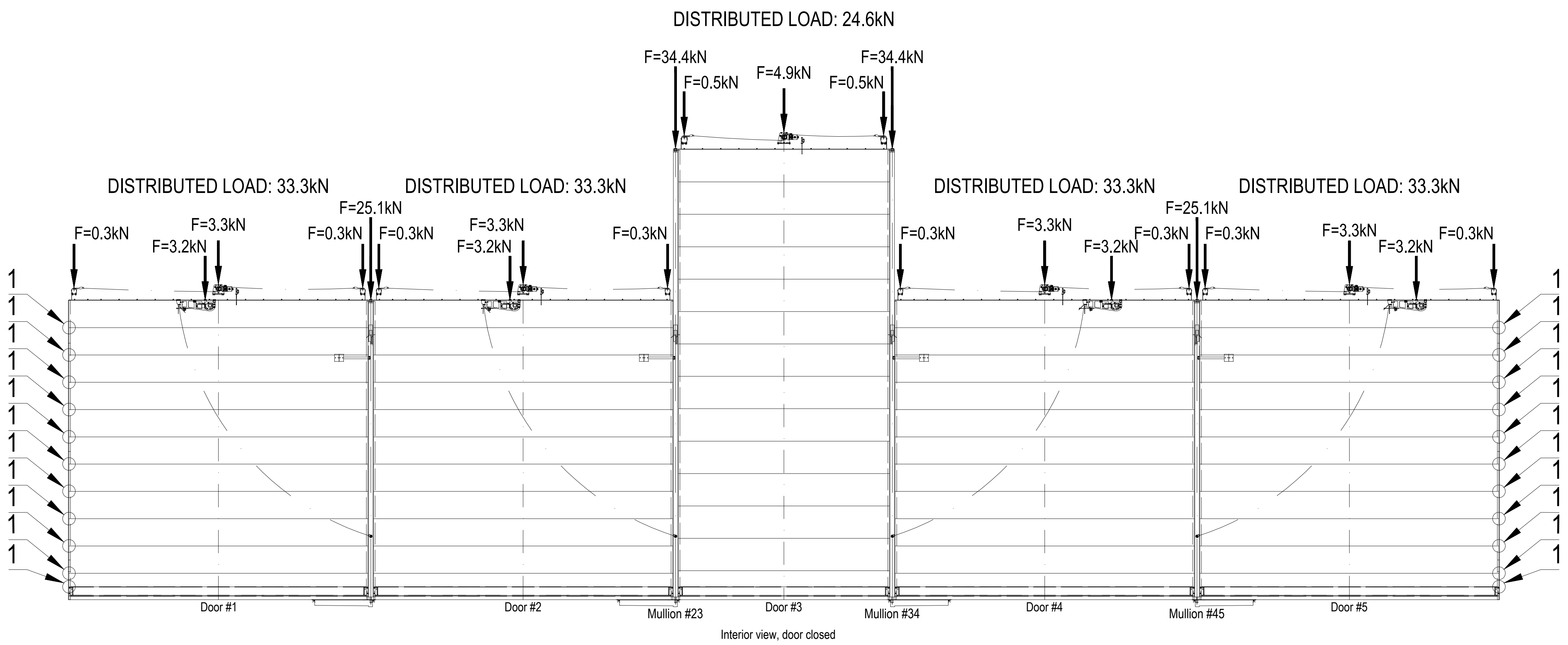
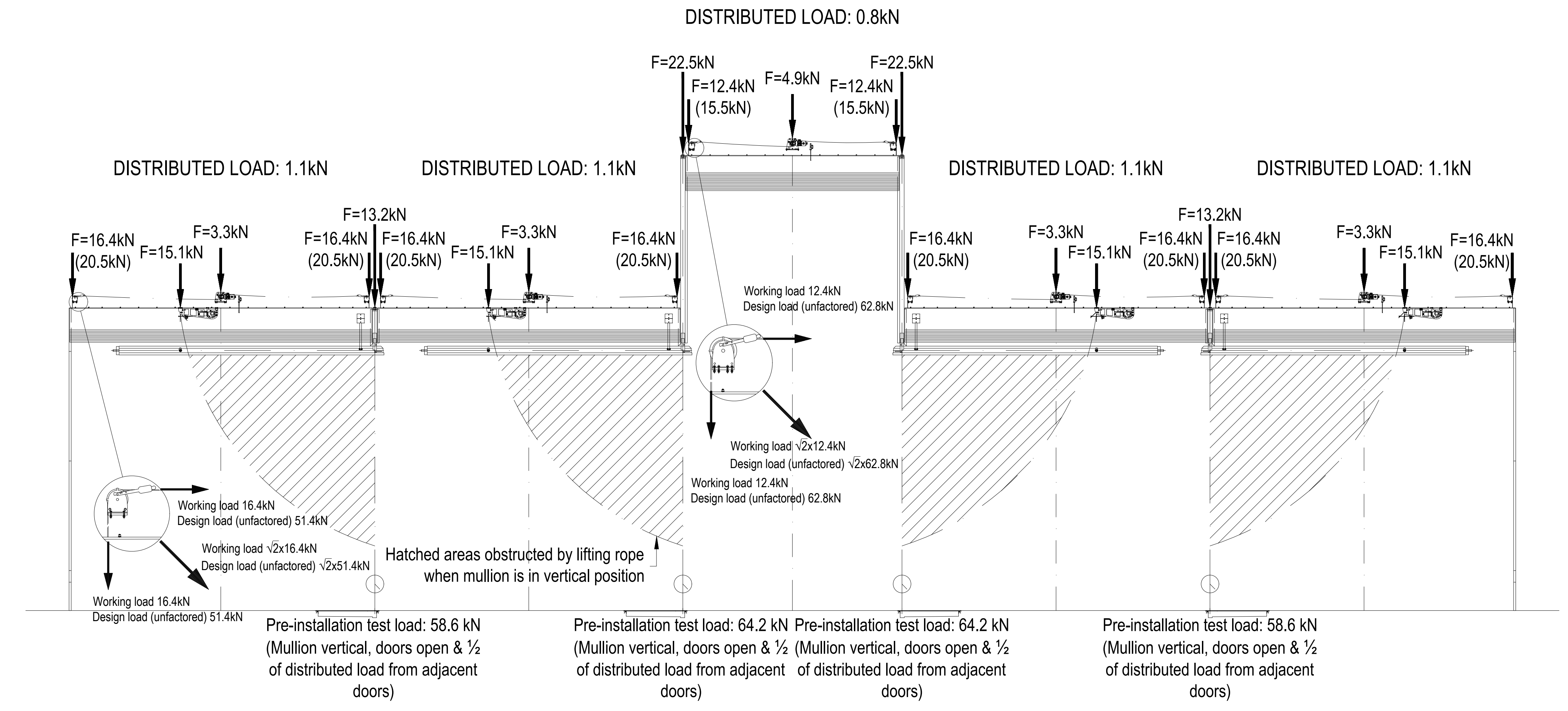
Revision		Date	By
A	Corrected mullion hoist positions, floor pits altered	2022-09-02	EB
B	New measurements, doors redrawn, recalculated	2022-10-12	EB

LOAD DIAGRAM

Nominal static vertical loads are indicated. Value within brackets include dynamic surload factor 1.25 for door leaf loads in open position.
Includes items provided by ASSA ABLOY only.

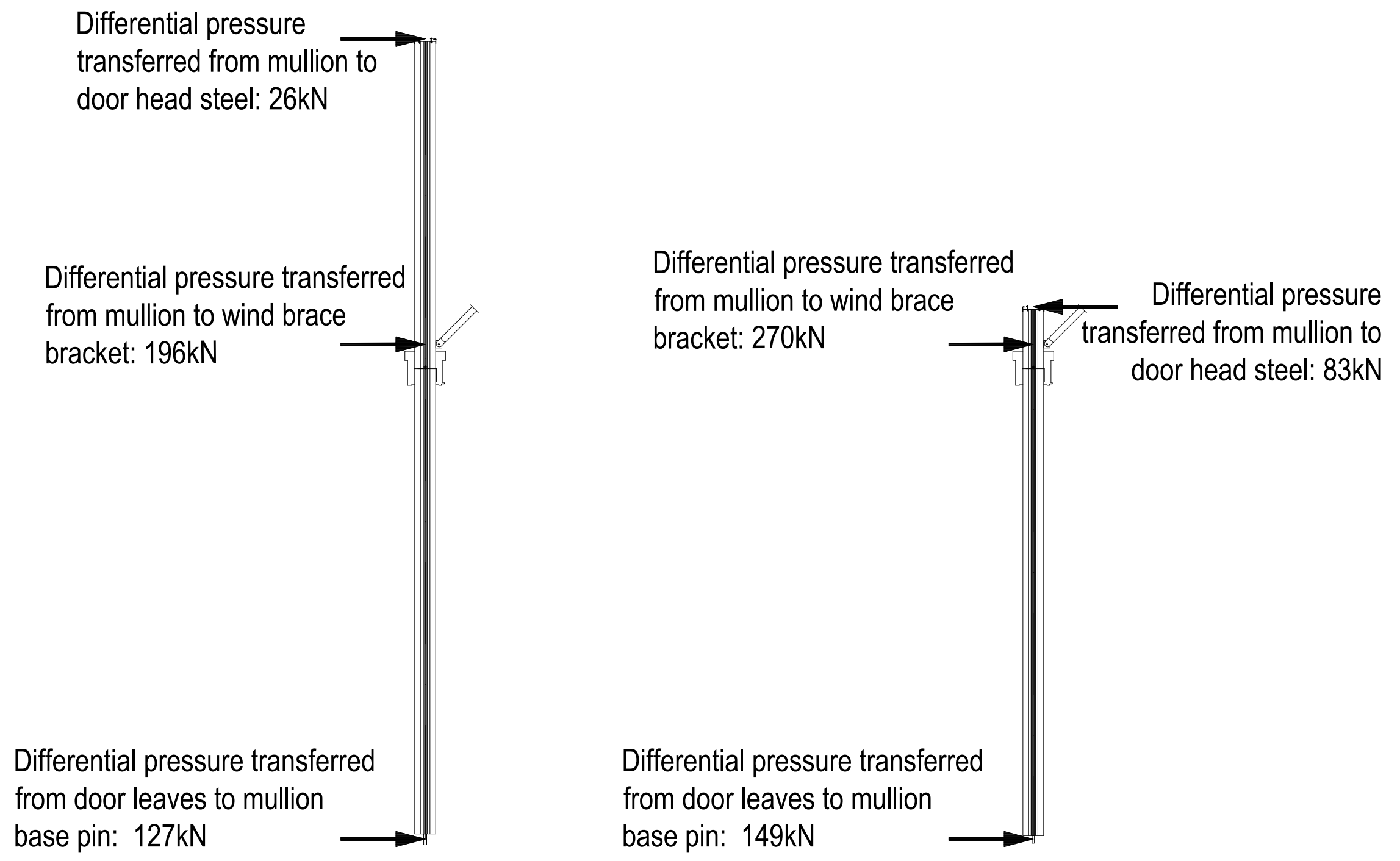

DISTRIBUTED LOAD: 24.6kN
F=34.4kN
F=0.5kN
F=4.9kN
F=0.5kN
F=34.4kN
DISTRIBUTED LOAD: 33.3kN
DISTRIBUTED LOAD: 33.3kN
DISTRIBUTED LOAD: 33.3kN
DISTRIBUTED LOAD: 33.3kN
F=0.3kN
F=3.2kN
F=3.3kN
F=25.1kN
F=0.3kN
F=0.3kN
F=3.3kN
F=3.2kN
F=0.3kN
F=0.3kN
F=3.3kN
F=3.2kN
F=25.1kN
F=0.3kN
F=0.3kN
F=3.3kN
F=3.2kN
F=0.3kN
1
Door #1
Door #2
Mullion #23
Door #3
Mullion #34
Door #4
Mullion #45
Door #5

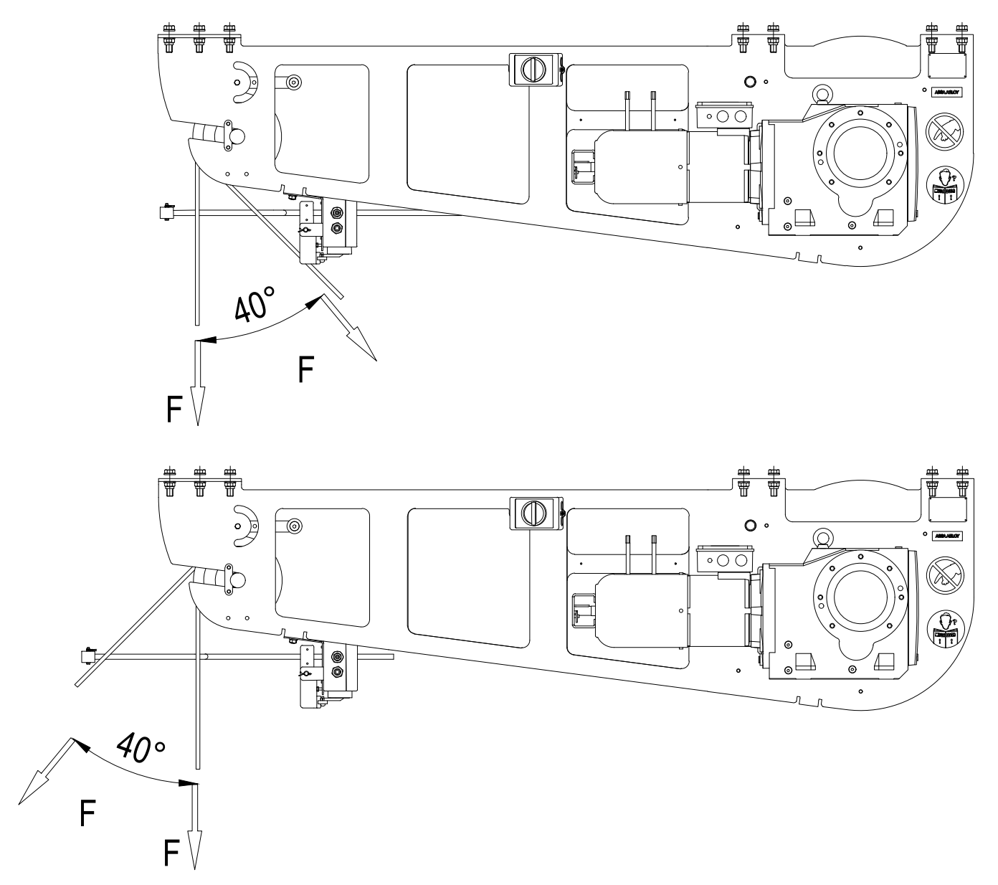
Interior view, door closed


DISTRIBUTED LOAD: 0.8kN
F=22.5kN
F=12.4kN (15.5kN)
F=4.9kN
F=12.4kN (15.5kN)
F=22.5kN
DISTRIBUTED LOAD: 1.1kN
DISTRIBUTED LOAD: 1.1kN
DISTRIBUTED LOAD: 1.1kN
DISTRIBUTED LOAD: 1.1kN
F=16.4kN (20.5kN)
F=15.1kN
F=3.3kN
F=13.2kN
F=16.4kN (20.5kN)
F=16.4kN (20.5kN)
F=15.1kN
F=3.3kN
F=16.4kN (20.5kN)
F=16.4kN (20.5kN)
F=3.3kN
F=15.1kN
F=13.2kN
F=16.4kN (20.5kN)
F=16.4kN (20.5kN)
F=3.3kN
F=15.1kN
F=16.4kN (20.5kN)
Working load 12.4kN
Design load (unfactored) 62.8kN
Working load √2x12.4kN
Design load (unfactored) √2x62.8kN
Working load 12.4kN
Design load (unfactored) 62.8kN
Working load 16.4kN
Design load (unfactored) 51.4kN
Working load √2x16.4kN
Design load (unfactored) √2x51.4kN
Working load 16.4kN
Design load (unfactored) 51.4kN
Hatched areas obstructed by lifting rope when mullion is in vertical position
Pre-installation test load: 58.6 kN (Mullion vertical, doors open & ½ of distributed load from adjacent doors)
Pre-installation test load: 64.2 kN (Mullion vertical, doors open & ½ of distributed load from adjacent doors)
Pre-installation test load: 64.2 kN (Mullion vertical, doors open & ½ of distributed load from adjacent doors)
Pre-installation test load: 58.6 kN (Mullion vertical, doors open & ½ of distributed load from adjacent doors)

Interior view, door open

Steel building company in cooperation with door installing company must make a pre-installation load test on the structure, where the mullions are located, in order to simulate the vertical deflection of the structure that occurs when doors are installed. This vertical deflection figure will be used to determine the correct installation level of the mullions. Required material/equipment and personnel by others than ASSA ABLOY.

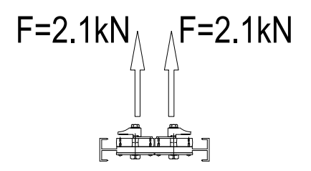

Differential pressure transferred from mullion to door head steel: 26kN
Differential pressure transferred from mullion to wind brace bracket: 196kN
Differential pressure transferred from door leaves to mullion base pin: 127kN
Differential pressure transferred from mullion to wind brace bracket: 270kN
Differential pressure transferred from mullion to door head steel: 83kN
Differential pressure transferred from door leaves to mullion base pin: 149kN



F = Design load (unfactored) 20.5kN

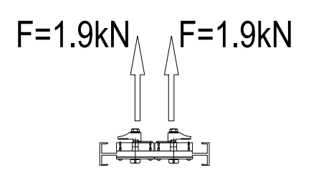
Force from mullion wire rope hoists
Scale 1:20

Load on top beam screws due to wind load and door weight.
Door #3

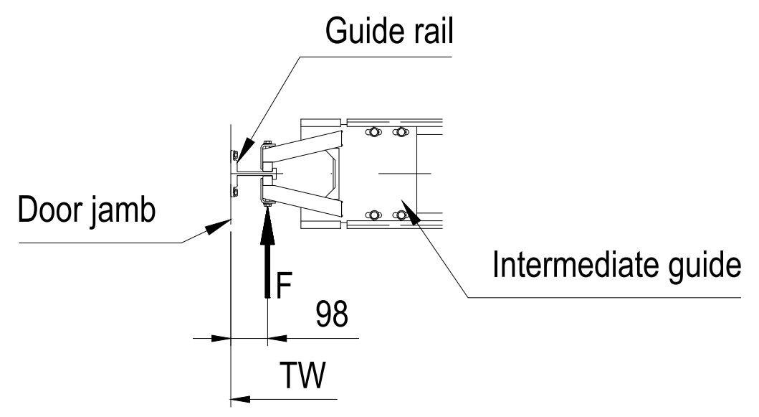

F=2.1kN
F=2.1kN

Section thru top beam
Scale 1:20

Load on top beam screws due to wind load and door weight.
Door #1, 2, 4 & 5

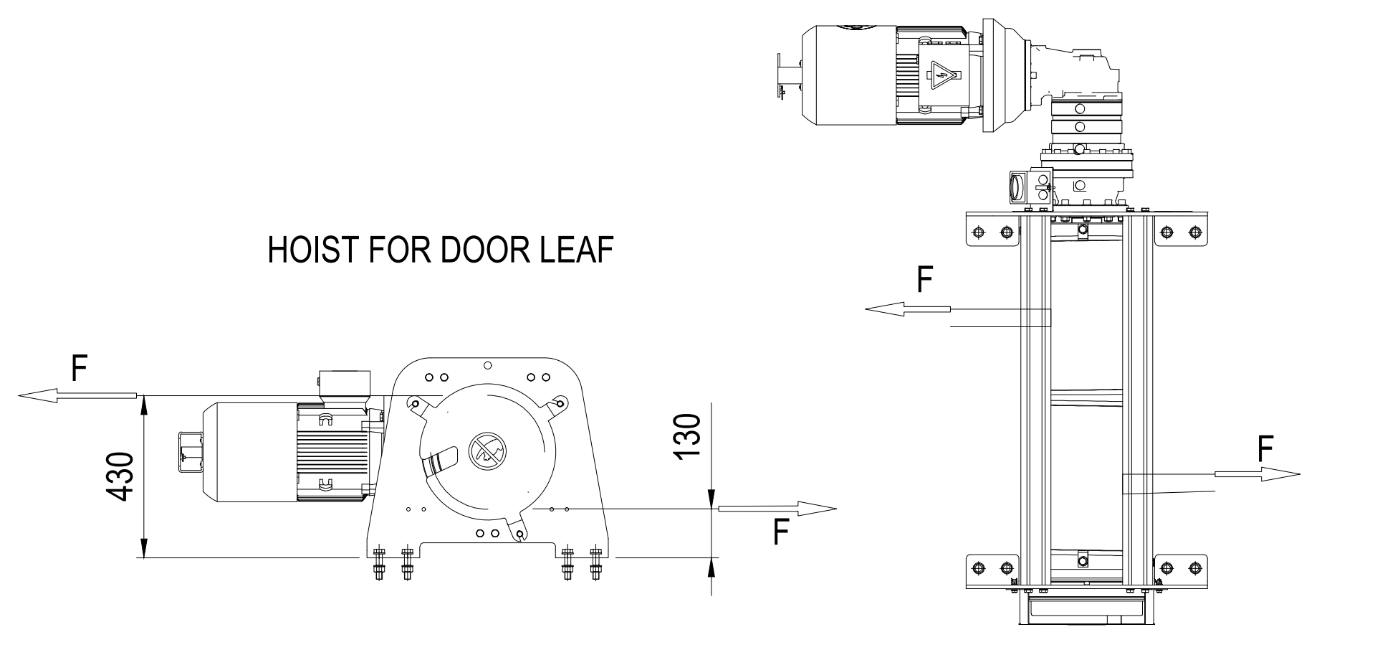

F=1.9kN
F=1.9kN

Section thru top beam
Scale 1:20


Guide rail
Door jamb
Intermediate guide
F
98
TW

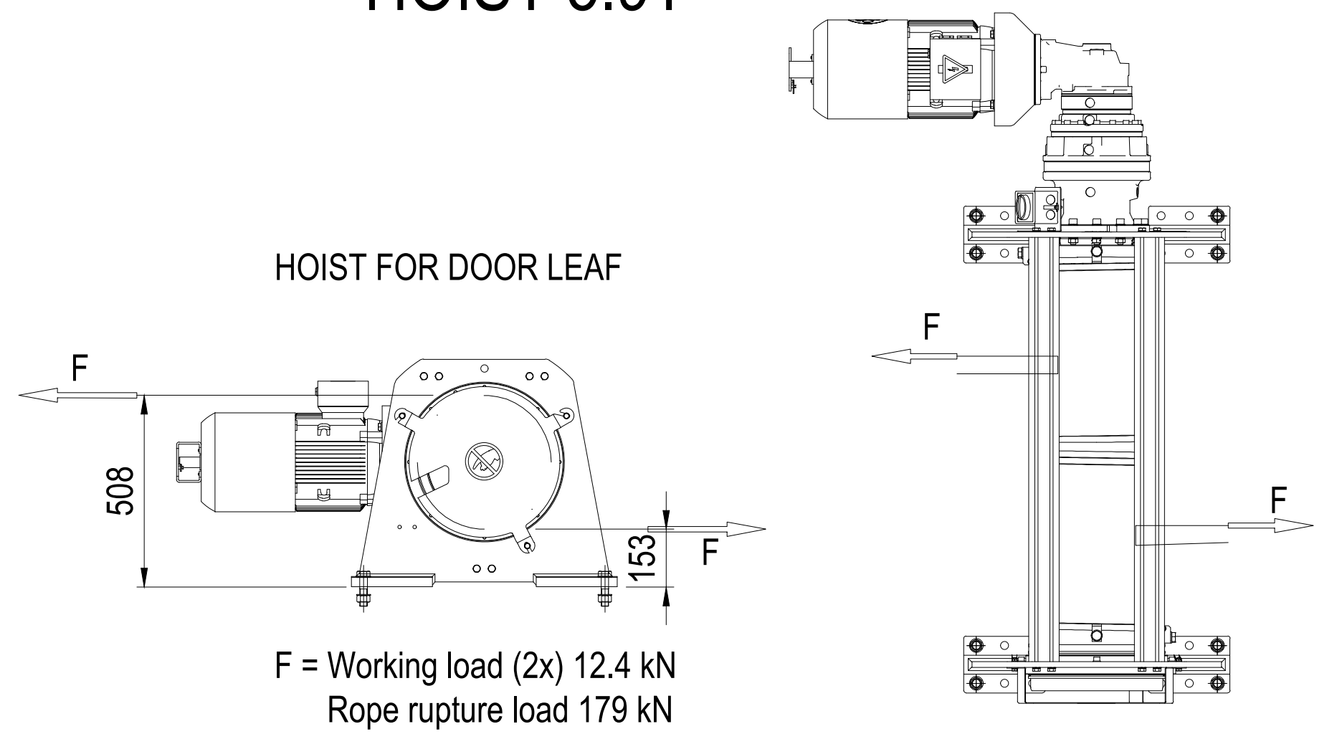
Force by differential pressure transferred to guide rail at each intermediate position.
(located at pos 1)
F=14.8 kN

HOIST 4.9T


HOIST FOR DOOR LEAF
F
F
430
130
F
F

F = Working load (2x) 16.4 kN
Rope rupture load 100 kN

HOIST 5.9T


HOIST FOR DOOR LEAF
F
508
153
F
F
F

F = Working load (2x) 12.4 kN
Rope rupture load 179 kN

NOT INCLUDED IN DOOR SUPPLY:

1. Design and supply of all door supporting structural components.
2. Design and supply of all closure between door and adjacent steel.
3. Design and supply of coverage, such as roofing or enclosure, for door components at door head.
4. Door jambs adjustable for overall door width and true to plumb.
5. Paint job after completed door installation.
6. Any frameworks being provided by ASSA ABLOY will be clearly indicated as "by ASSA ABLOY".

REQUIREMENTS ON BUILDING:

7. Door head support must be straight within 1/600 of span in the in/out direction and free from protrusions of any kind.
8. For a single door leaf, door head support should be horizontal and straight within 1/600 of span at distributed full door load.
9. Door side jambs must be smooth, straight and parallel, out of plumb must not exceed 2mm/m (total deviation at full length 5 mm) in any direction.
10. Any footings at bottom of jamb columns must not interfere with installation of guide rails at floor level.
11. At time of installation, finished floor, free from obstructions, out to a distance of 2 m on each side of door centreline must be available.
12. Bracket/attachment plate for hoist should be flat within 5mm/m.

DOOR PROPERTIES:

13. The door is designed to withstand a differential pressure of 1.2kPa.
14. Mullions and mullion pits are designed to allow for 128mm of downward deflection and 128mm of upward deflection due to external loads such as wind.
15. Guide rails, top- and intermediate beams in extruded aluminium with natural finish.
16. Environmental category: Megadoor standard
17. Bottom beam in carbon steel painted: Light grey (RAL7047)
18. Colored clamp strips on both sides
19. The door fabric is a UV stabilized, flame retardant, self-extinguishing, PVC coated polyester fabric with fire technical properties Euroclass B-s2, d0 (according to EN 13501-1:2007).
    Colour of fabric: TBC
    RED (NCS S2070-R, RAL 3001)
    BLUE (NCS S3560-R80B, RAL 5005)
    GREY (NCS 3500, RAL 7004)
    WHITE (NCS 0500, RAL 9016)
    TAN (NCS 2010-Y40R, RAL 1001)
    WHITE ALUMINUM (RAL 9006)
    GREEN (NCS 8010-G10Y, RAL 6009)
    ANTHRACITE GREY (NCS 8005-B20G, RAL 7016)
    (Please specify)

ELECTRICAL:

20. One main control panel provided of PLC type.
21. One additional screen.
22. Control mode : Impulse up / Constant contact down.
23. Cables are included.
24. Back-up power socket & switch.
25. Audible & visual alarm.
26. Approx. size of main control panel: WxHxD (500x600x250 mm),
27. Location of main control panel: Please specify location.
28. Aux. Voltage 24VDC
29. Power supply: 400V /3Ph /50Hz
30. Power: 41.3kW
31. Nom. current: 97.7A
32. Start. current: 214.2A
33. (max. allowed voltage drop at start: 5%)

WIRE ROPE HOIST UNIT:

LIDAN 4.9T2/27(4x)
2x Ø12mm wire ropes. Rupture load 100kN/rope.

LIDAN 5.9T2/27(1x)
2x Ø16mm wire ropes. Rupture load 179kN/rope.

MULLION HOIST UNIT:

LIDAN 1.75T/1(4x)
2x Ø10mm wire ropes. Rupture load 69.8kN/rope.

DOOR IDENTITY NUMBER: 84800001

INTERFACE, SECTIONS ref. to drwng. no. 77445

INTERFACE, PLAN VIEWS ref. to drwng. no. 77444

INTERFACE, ELEVATION ref. to drwng. no. 77443

DESIGNED EB	SIZE A1F	SCALE 1:200	GMR Aero Technic Hangar1 India Megadoor 84000x23000/14000 Load diagram & Notes
CODE 84800	If not otherwise stated: Tolerances ISO 2768-m EN ISO 13920-B Cutting class ISO 9013-442 Weld class ISO 5817-C Surface finish Smooth raw surface		PLOT DATE
ITEM NO			DRAWING NO 77446
DATE 2022-08-24			REVISION B
ASSA ABLOY			

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